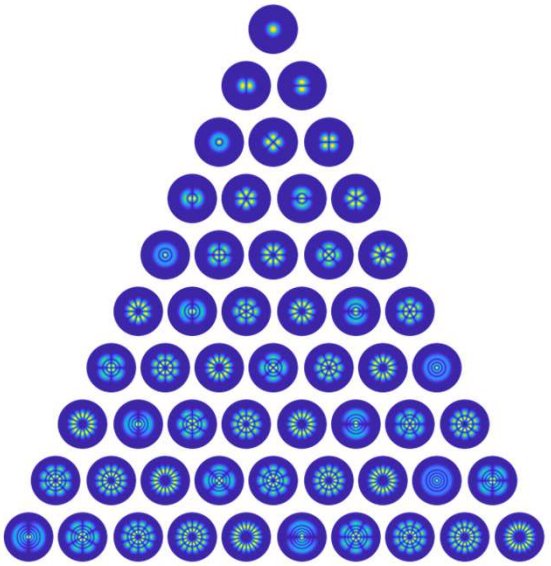
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2024/25

Lecture 2



Prof. Dr.-Ing. Georg Rademacher

Organizational matter :



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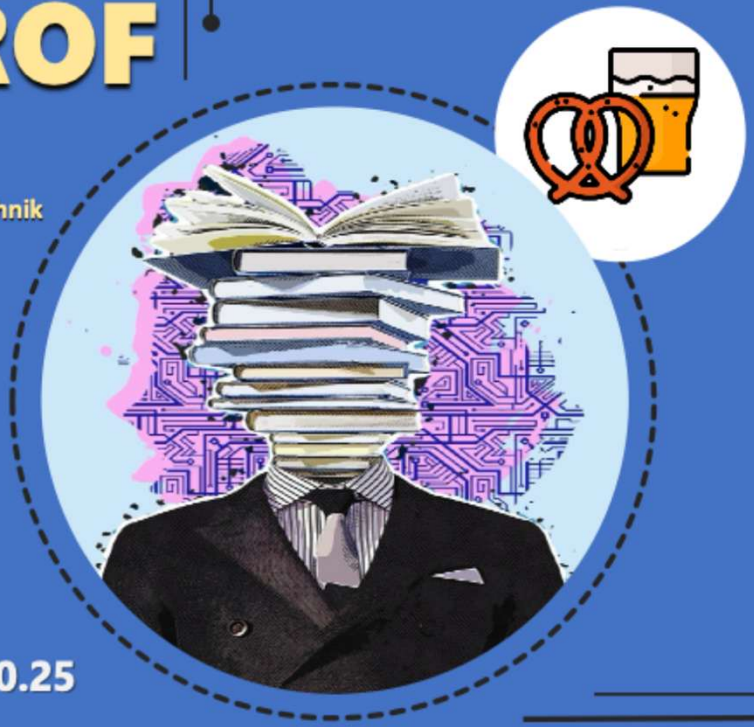
- This module contains a lecture and an exercise
- Typically, the lecture is held Thursdays at 2 pm in room (V 47-06)
- The exercise is on Tuesdays at 2 pm in this room (V 47-06) and will be held by Julian Schneck

- There won’t be a course reader (skript) but we will upload the slides from this lectures
- We recommend Govind P. Agrawal, Fiber Optic Communications Systems

MEET YOUR PROF

BIER- & BREZEL-ABEND*

Für Erstis des Bachelor-Studiengangs Elektrotechnik und Informationstechnik



13. November

17:30 Uhr

UG Pfaffenwaldring 47

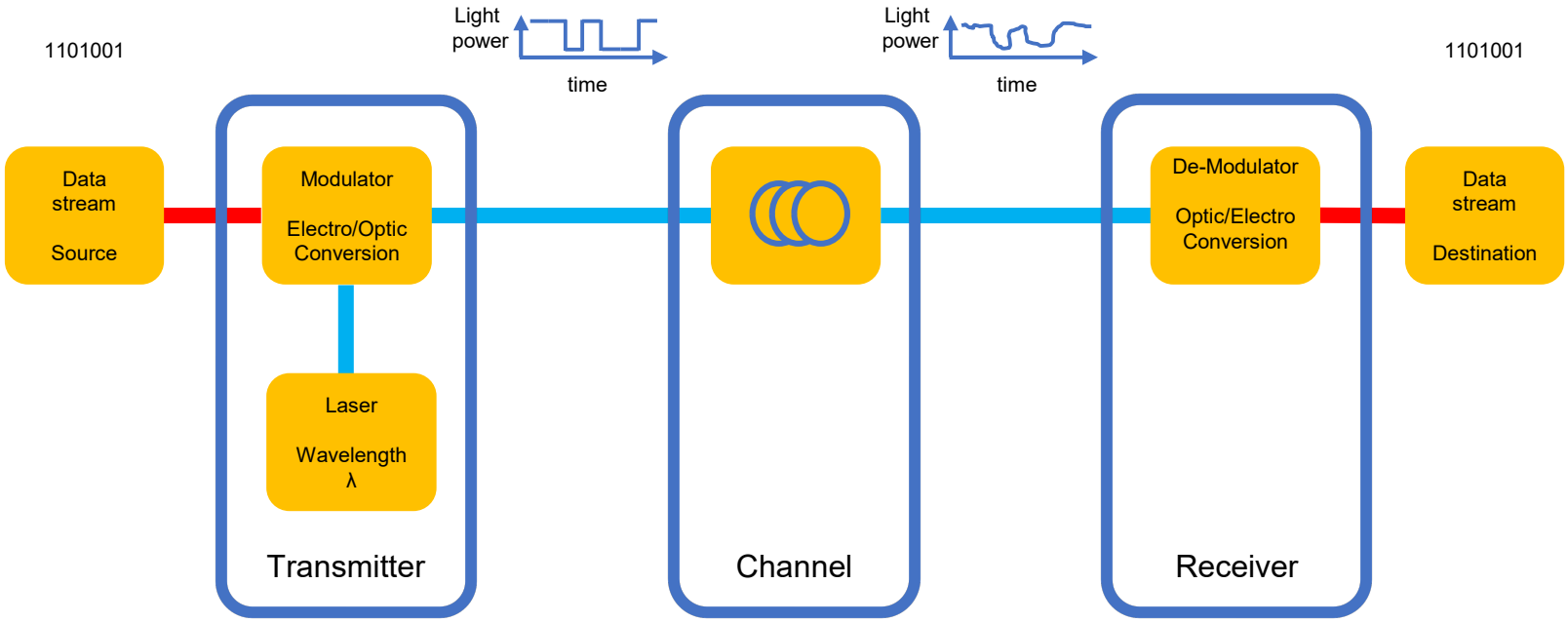
Anmeldung erfolgt über C@MPUS bis 16.10.25

Komm vorbei und lerne Deine Professoren und Professorinnen in lockerer Atmosphäre kennen!

* Bier, Brezeln sowie alkoholfreie Getränke sind gratis, solange der Vorrat reicht!

Principle of an optical fiber transmission link

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Germany



Plane waves

$$\varepsilon = \varepsilon_0 \cdot \varepsilon_r \quad , \quad \varepsilon_r = n^2$$

$$\nabla \times \underline{\vec{E}} = -j\omega\mu_0 \underline{\vec{H}}$$

$$\nabla \times \underline{\vec{H}} = j\omega\varepsilon \underline{\vec{E}} = j\omega\varepsilon_0 n^2 \underline{\vec{E}}$$

Plane waves

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Germany

$$\nabla \times (\nabla \times \underline{\vec{E}}) = k_0^2 n^2 \underline{\vec{E}} \quad , \quad k_0 = \omega \sqrt{\mu_0 \varepsilon_0}$$

$$\nabla \times (\nabla \times \underline{\vec{E}}) = \nabla \underbrace{(\nabla \cdot \underline{\vec{E}})}_{=0} - \nabla^2 \underline{\vec{E}} = -\nabla^2 \underline{\vec{E}}$$

Plane waves

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$$\nabla^2 \underline{\vec{E}} + k_0^2 n^2 \underline{\vec{E}} = 0$$

$$\nabla^2 \underline{\vec{H}} + k_0^2 n^2 \underline{\vec{H}} = 0$$

$$\nabla^2 : \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\underline{\vec{E}} = \underline{\vec{E}}_0 \exp(-j \cdot \vec{k} \cdot \vec{r})$$

$$= \underline{\vec{E}}_0 \exp(-j(k_x \cdot x + k_y \cdot y + k_z \cdot z))$$

$\vec{k}$: wave vector

Plane waves

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$$\underline{\vec{H}} = \frac{\vec{k} \times \underline{\vec{E}}}{\omega \mu_0} = \underline{\vec{H}}_0 \exp(-j \vec{k} \cdot \vec{r})$$

$$\underline{\vec{E}} = -\frac{\vec{k} \times \underline{\vec{H}}}{\omega \varepsilon_0 n^2}$$

$$\vec{S} = \frac{1}{2} \left(\underline{\vec{E}} \times \underline{\vec{H}}^* \right)$$

energy flux

Plane waves

only propagates into $\vec{z}$ $\rightarrow$ $\vec{k} = \vec{e}_z \, k_0 \cdot n$

$$\underline{\vec{E}} = \underline{\vec{E}}_0 \exp(-j k_0 n z)$$

$$\underline{\vec{E}}_0 = \begin{pmatrix} \underline{a}_x \\ \underline{a}_y \end{pmatrix} \underline{E}_0$$

$\begin{pmatrix} \underline{a}_x \\ \underline{a}_y \end{pmatrix}$: Jones-Vector , $|\underline{a}_x|^2 + |\underline{a}_y|^2 = 1$

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$\underline{a}_x = 1$, $\underline{a}_y = 0$: linearly polarized in x-direction

$$\underline{E}_x = \underline{E}_0 \exp(-j k_0 n z) \quad , \quad \underline{E}_y = \underline{E}_z = 0$$

$$\underline{H}_y = \underline{H}_0 \exp(-j k_0 n z) \quad , \quad \underline{H}_x = \underline{H}_z = 0$$

$$\frac{\underline{E}_x}{\underline{H}_y} = Z_F = \frac{\underline{E}_0}{\underline{H}_0} = \frac{1}{n}\sqrt{\frac{\mu_0}{\varepsilon_0}} = \frac{120\,\pi\,\Omega}{n}$$



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Concept Test

Concept test

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The electric field vector of a plane wave has only a component in z-direction. Which directions of the magnetic field and the Poynting vector are not possible?

A $\vec{H}$: -y, $\vec{S}$: x

B $\vec{H}$: x, $\vec{S}$: y

C $\vec{H}$: x, $\vec{S}$: -y

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Plane waves

$$E_x(t) = \mathrm{Re}\left\{ \underline{E}_0 \exp\left(j\left[\omega t - \beta z\right]\right)\right\}$$

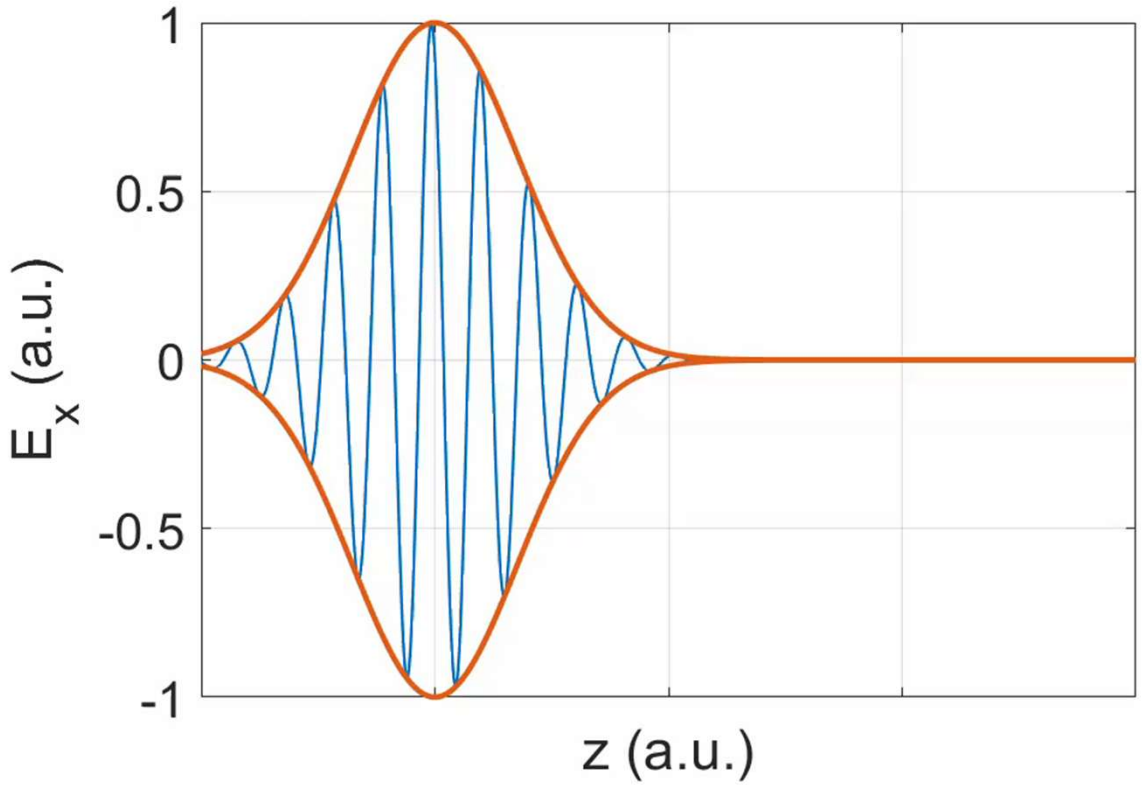
$\beta = k_0 \cdot n$: Propagation Constant

, if E_0 real:

$\rightarrow$ $E_x(t) = E_0 \cdot \cos(\rho)$, $\rho = \omega t - \beta z$: Phase of wave

$v_{ph} = \frac{dz}{dt} = \frac{\omega}{\beta}$, $\omega t - \beta z = \text{const.}$

Chromatic Dispersion



Chromatic Dispersion

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$$v_{gr} = \frac{d\omega}{d\beta} \quad , \quad \beta = k_0 \cdot n \quad , \quad k_0 = \frac{\omega}{c} \quad , \quad c = \frac{1}{\sqrt{\mu_0 \cdot \varepsilon_0}}$$

$$v_{ph} = \frac{k_0 \cdot c}{k_0 \cdot n} = \frac{c}{n}$$

$$v_{gr} = \left(\frac{d(k_0 \cdot n(\omega))}{d\omega} \right)^{-1} = \frac{c}{\frac{d(\omega\, n(\omega))}{d\omega}} = \frac{c}{NV}$$

$$c \approx 3 \cdot 10^8 \, \frac{m}{s}$$

Chromatic Dispersion

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$$N = \frac{d(\omega \cdot n(\omega))}{d\omega} = n(\omega) + \omega \frac{dn(\omega)}{d\omega} = n(\lambda) - \lambda \frac{dn(\lambda)}{d\lambda}$$

$$n \neq N$$

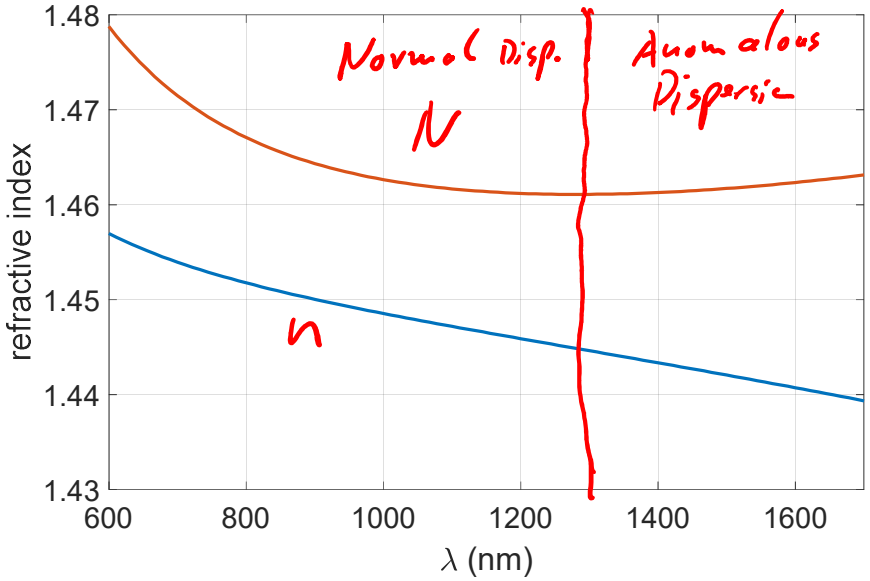
$$t_{gr} = \frac{L}{v_{gr}} = L \cdot \tilde{\tau}$$

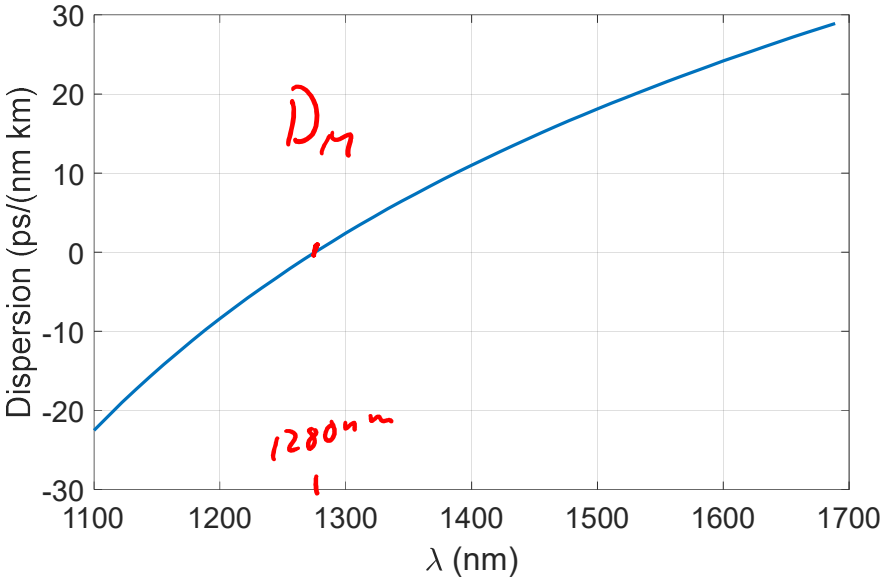
$$\tilde{\tau} = \frac{1}{v_{gr}} = \frac{N}{c}$$

$$\frac{d\tilde{\tau}}{d\lambda} = \frac{1}{c} \cdot \frac{dN}{d\lambda} = -\frac{\lambda}{c} \frac{d^2 n(\lambda)}{d\lambda^2} = D_M$$

Chromatic Dispersion

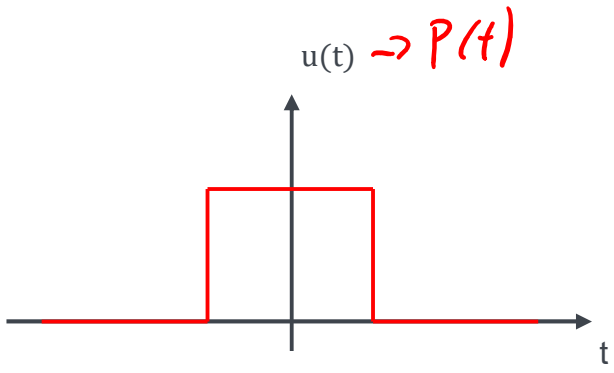
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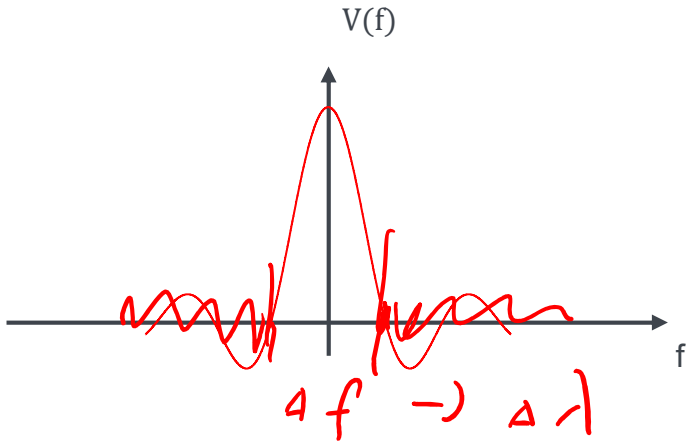




Dispersion

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$$\Delta t \approx L \cdot \Delta\lambda \cdot D_M$$



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Concept Test

Concept test

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What is the delay time difference Δt due to chromatic dispersion with $\Delta\lambda = 0.2\ \mathrm{nm}$, $L = 50\ \mathrm{km}$, and $D_\mathrm{M} = 17\ \frac{\mathrm{ps}}{\mathrm{nm\ km}}$?

A 170 ps

B 170 ns

C 170 µs

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Refractive index with Sellmeier equations

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$$\varepsilon_r - 1 = n^2 - 1 = \frac{B_1}{\omega_1^2 - \omega^2} \qquad : |\omega - \omega_1| < 0{,}1\,\mu m$$

$$n^2(\omega) - 1 = \sum_i \frac{B_i}{\omega_i^2 - \omega^2} \qquad : \; 3\text{-}5 \text{ terms}$$

$$0{,}8\,\mu m < \lambda < 1{,}6\,\mu m \qquad : \; 2 \text{ Resonances}$$

Refractive index with Sellmeier equations

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$$n^2 - 1 = \frac{B_1}{\omega_1^2 - \omega^2} + \frac{B_2}{\omega_2^2 - \omega^2}$$

, $\omega_2 \ll \omega$, $\lambda_2 \approx 10\,\mu m$

$$\rightarrow \quad n^2 - 1 = \frac{B_1}{\omega_1^2 - \omega^2} + \frac{B_2}{\omega^2}$$

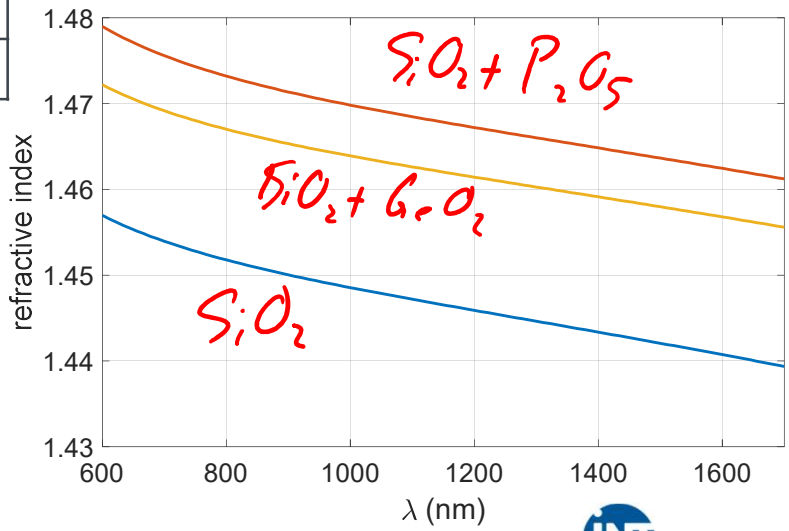
$\omega_e = \omega_1$, ω_d, ω_L $\rightarrow$ $B_1 = \omega_e \cdot \omega_d$, $B_2 = \omega_L^2$

$$\rightarrow \quad n^2 = 1 + \frac{\omega_e \cdot \omega_d}{\omega_e^2 - \omega^2} - \frac{\omega_L^2}{\omega^2}$$

Refractive index with Sellmeier equations

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Molar composition in %					f_e	f_d	f_L
SiO_2	B_2O_3	GeO_2	P_2O_5	Na_2O	in 10^{14} Hz		
100					32,12	35,31	0,31
86,7	13,3				31,27	34,20	0,33
86,5		13,5			30,21	34,99	0,29
90			10		32,05	36,61	0,29
58	19			23	30,53	39,34	0,31
18	58			24	30,98	39,27	0,36



Attenuation

$$n = n' - j n''$$

$$\beta = k_0 n' \quad , \quad \alpha = k_0 \cdot n''$$

$$\underline{E}_x = E_{x0} \exp(-j\beta z - \alpha z)$$

$$\alpha\,[\mathrm{dB/km}] = \frac{10 \cdot \log\left(\frac{P_0}{P_0 \cdot \exp(-2\alpha L)}\right)}{L} = \alpha \cdot 8.69\ \mathrm{dB}$$

1. Scattering (Rayleigh-scattering)

2. Absorption

Attenuation

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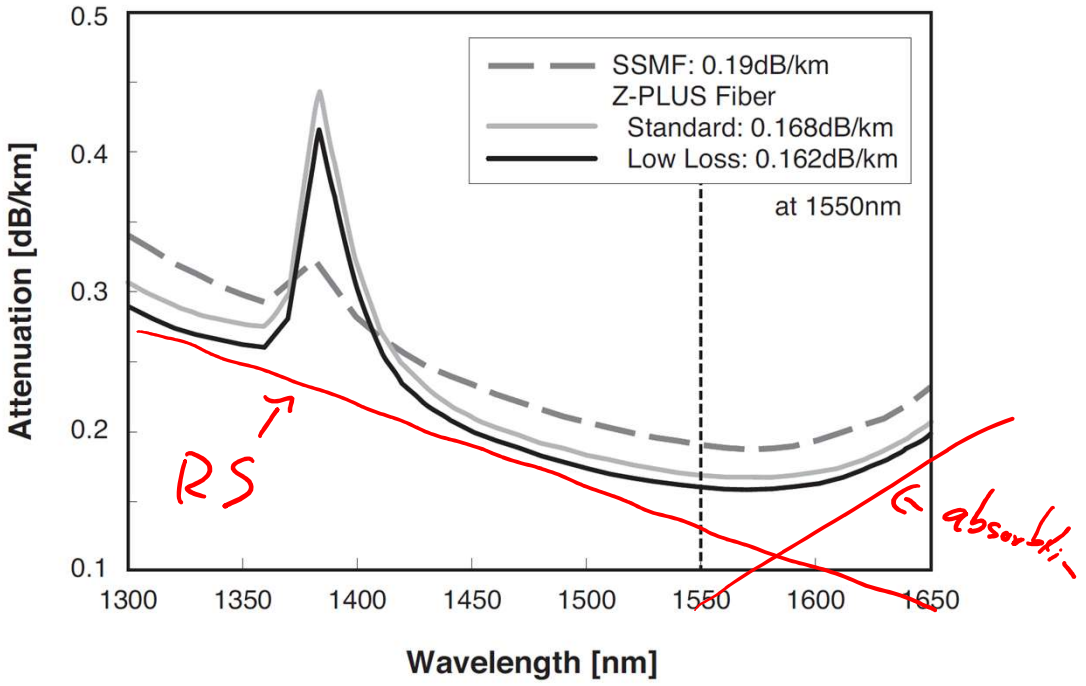
$\alpha_R = \frac{C}{\lambda^4}$

ultraviolett ($\lambda < 0{,}4\mu m$)

vibrational resc. : $\lambda > 7\mu m$

OH-ions $\approx 2.73\,\mu m$

1st Harmonic : $1{,}39\mu$



Yamamoto et al, Low-Loss and Low-Nonlinearity Pure-Silica-Core Fiber for Large Capacity Transmission, SEI Technical review, 2013



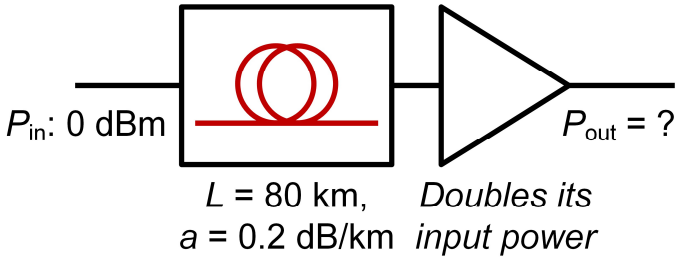
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Concept Test

Concept test

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What is the output power of the following optical channel?



A 13 dBm

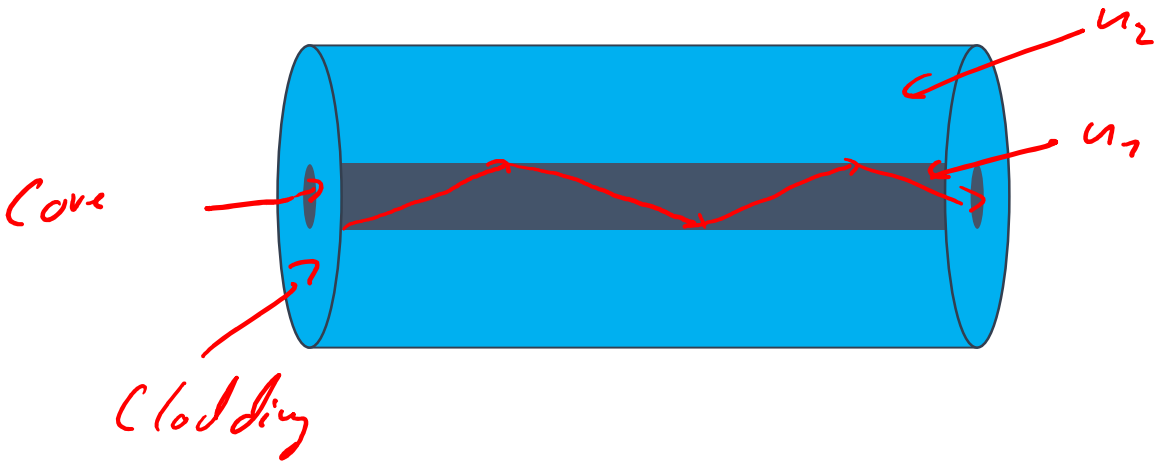
B -8 dBm

C -13 dBm

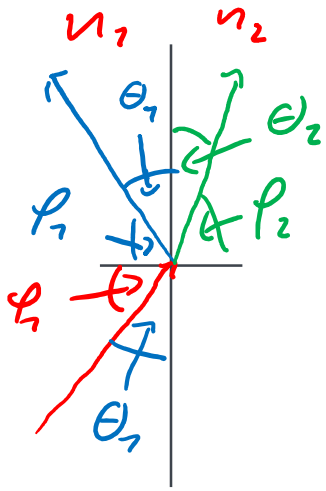
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Plane waves on dielectric interfaces

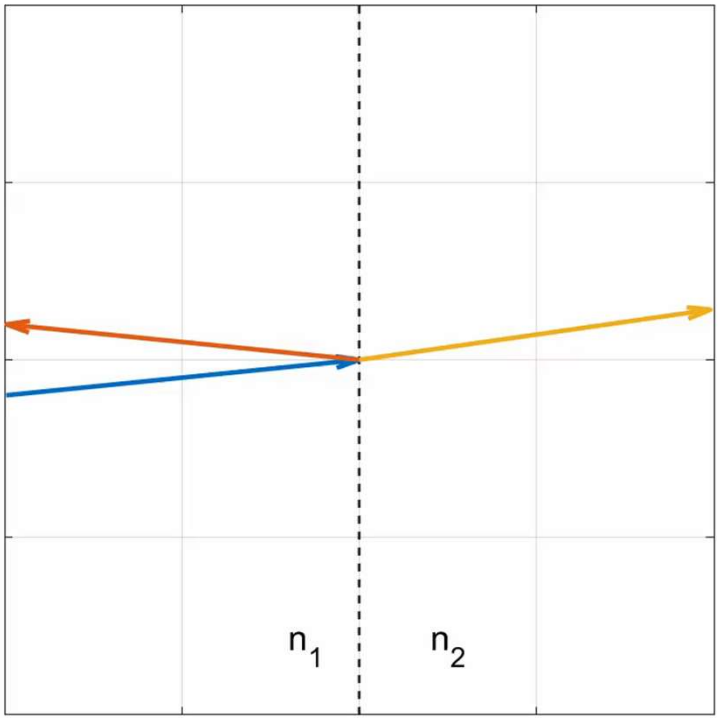
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Snell's law

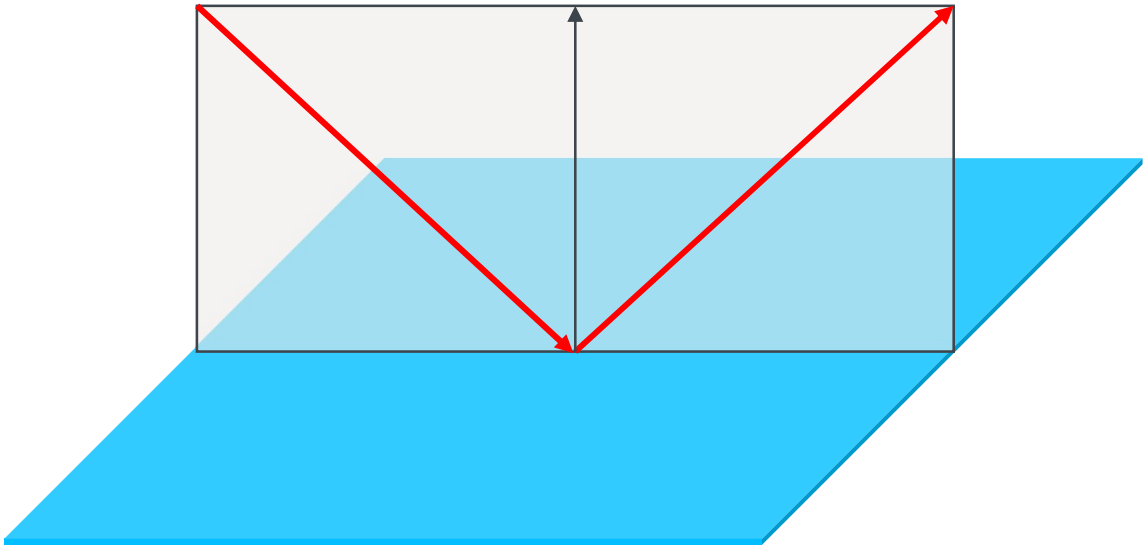


$$\frac{n_1}{n_2} = \frac{\cos(\theta_2)}{\cos(\theta_1)} = \frac{\sin(\rho_2)}{\sin(\rho_1)}$$

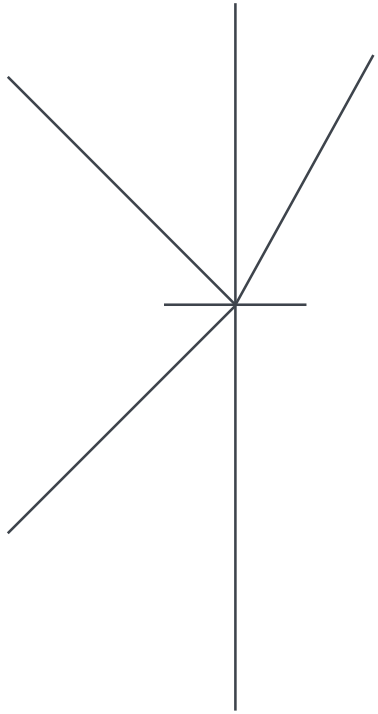


Plane of insidence

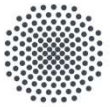
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Plane waves on dielectric interfaces



Plane waves on dielectric interfaces



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Reflectivity of plane waves on dielectric interfaces



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Plane waves on dielectric interfaces

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Thank you!

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